

SCALEMODE-NEURALSURV: MULTI-MODE SUBSPACE LINEARIZATION AND COVARIATE SCALE-TILTING FOR CONTINUOUS-TIME BAYESIAN SURVIVAL ANALYSIS

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ABSTRACT

Deep survival models deployed in data-scarce medical regimes must deliver both accurate risk stratification and well-calibrated epistemic uncertainty. State-of-the-art continuous-time Bayesian neural network (BNN) survival models such as NeuralSurv suffer from two structural constraints: (1) single maximum-a-posteriori (MAP) point-linearization collapses non-convex parameter posteriors, yielding overconfident predictive bands, and (2) sigmoidal hazard modulation $\sigma(g(t, x)) \in (0, 1)$ imposes a strict ceiling at the baseline rate $\lambda_0(t)$, preventing risk amplification for high-risk subgroups. We introduce **ScaleMode-NeuralSurv**, a continuous-time Bayesian survival framework that addresses both bottlenecks. It combines *multi-center subspace linearization* through anchored randomized priors $a_k \sim \mathcal{N}(0, \sigma_a^2 I_m)$, which capture multimodal functional diversity, with an unconstrained *covariate scale-tilting* factor $e^{w^\top x}$ that bypasses the attenuation ceiling. Crucially, continuous-time Pólya-Gamma and marked Poisson process augmentations preserve closed-form Coordinate Ascent Variational Inference (CAVI) updates at linear $\mathcal{O}(Km)$ complexity. Across eight real-world survival benchmarks, ScaleMode-NeuralSurv improves the aggregate Antolini C-index from 0.593 to 0.646, reduces the IPCW Integrated Brier Score from 0.209 to 0.194, and lifts empirical epistemic coverage ($PICP_{km}$) from 0.539 to 0.605, outperforming twelve deep survival baselines. Under an identical-splits protocol that additionally includes Deep Cox Mixtures, SurvUnc, and conformalized survival baselines, it retains the best aggregate C-index (0.648) and IPCW-IBS (0.187), and its gains persist to full-scale cohorts (+0.100 C-index on full METABRIC; +0.042 on full SUPPORT, $N=8,873$) with numerically stable CAVI. Extensive ablations reveal that scale-tilting is the dominant driver—amplifying hazard by up to $24.9\times$ on high-risk subgroups, where a KL analysis confirms the local quadratic scale-tilt bound stays tight ($< 10^{-2}$ nats)—while a link-function swap shows the bounded sigmoid is what keeps the multi-mode posterior calibrated.

1 INTRODUCTION

In biomedical decision-making, modeling time-to-event outcomes—survival analysis—is essential for risk stratification, prognosis, and therapeutic selection (Wang et al., 2017). Unlike standard regression, time-to-event data exhibit right-censoring, where event times are only partially observed. To be useful in clinical decision support, deep survival models must deliver not only high discriminative performance but also well-calibrated probabilistic predictions with trustworthy epistemic uncertainty bounds (Kendall & Gal, 2017; Lakshminarayanan et al., 2016). Overconfident or miscalibrated survival predictions can jeopardize patient care by underestimating mortality or recurrence risk (Chapfuwa et al., 2019; Goldstein et al., 2020; Avati et al., 2018; Haider et al., 2018). Consequently, recent machine-learning-for-health research emphasizes calibration and robust uncertainty estimation (Jain et al., 2026; Nagpal et al., 2021; Liu et al., 2025).

Continuous-time Bayesian neural network (BNN) frameworks such as NeuralSurv (Monod et al., 2025) have emerged as a powerful paradigm for survival analysis in data-scarce regimes. Neural-

Surv achieves non-parametric, continuous-time modeling through exact Pólya-Gamma (Fu & Lu, 2023; Wenzel et al., 2018) and marked Poisson process data augmentations. By locally linearizing the network around its MAP parameters, it induces Gaussian conjugacy, enabling closed-form Coordinate Ascent Variational Inference (CAVI) updates that scale efficiently (Monod et al., 2024; Kingma & Ba, 2014).

Despite these advantages, continuous-time BNN survival models suffer from two structural constraints:

1. **Underestimated Epistemic Uncertainty (Limitation 1).** Single-MAP linearization—a local tangent-space / Neural Tangent Kernel approximation around one parameter mode—collapses the non-convex, multimodal parameter posterior into a single local Gaussian process. In data-scarce clinical regimes this severely underestimates functional uncertainty, so empirical coverage (e.g., Kaplan-Meier prediction interval coverage $PICP_{km}$) falls well below its nominal level (0.95).
2. **Hazard Attenuation Ceiling (Limitation 2).** Sigmoidal hazard modulation writes $\lambda(t|x) = \lambda_0(t) \sigma(g(t, x))$ with $\sigma(\cdot) \in (0, 1)$, so the instantaneous risk of any subject is strictly capped at or below the prior baseline $\lambda_0(t)$. The model therefore cannot amplify hazard for high-risk subgroups without globally rescaling the baseline.

To address both bottlenecks while preserving continuous-time augmentation and closed-form CAVI, we propose **ScaleMode-NeuralSurv**, a two-fold architectural advance:

- **Multi-Center Subspace Linearization via Anchored Ensembling.** Instead of linearizing around a single global MAP point, we build a mixture of K local tangent-space approximations centered at diverse modes $\{\theta_{MAP}^{(k)}\}_{k=1}^K$. To enforce functional diversity and prevent mode collapse, each mode is fit under an anchored randomized prior mean $a_k \sim \mathcal{N}(0, \sigma_a^2 I_m)$ that pulls experts into distinct parameter subspaces.
- **Scale-Tilted Hazard Representation.** On top of each fitted mode we add an unconstrained linear covariate scale-tilting factor $e^{w_k^\top x}$ with Gaussian prior $w_k \sim \mathcal{N}(0, \sigma_w^2 I_p)$, estimated by a conjugate proportional-hazards recalibration. Each mode’s hazard $\lambda^{(k)}(t|x) = \lambda_0(t) e^{w_k^\top x} \sigma(g_{lin}^{(k)}(t, x; \theta_k))$ permits unbounded multiplicative amplification ($e^{w_k^\top x} > 1$), breaking the sigmoidal ceiling, and the K modes are pooled uniformly at prediction time.

We evaluate ScaleMode-NeuralSurv across eight real-world survival benchmarks (Curtis et al., 2012) under 5-fold cross-validation in the data-scarce regime ($N = 125$), improving the average Antolini C-index from 0.593 to 0.646 (+0.053), reducing IPCW Integrated Brier Score (IBS) from 0.209 to 0.194, and raising epistemic coverage $PICP_{km}$ from 0.539 to 0.605. Our ablations position covariate scale-tilting $e^{w^\top x}$ as the *primary* driver of discriminative accuracy—lifting the maximum instantaneous hazard for high-risk subjects by up to $24.9\times$ over the baseline ceiling ($\Delta IBS = -0.061$ on the high-risk tier)—with multi-center linearization as a complementary mechanism that supplies multimodal parameter coverage without breaking continuous-time CAVI conjugacy. For rigor we re-train every baseline on the *identical* $N=125$ splits and add uncertainty-focused competitors—Deep Cox Mixtures (Nagpal et al., 2021), SurvUnc (Liu et al., 2025), and conformalized survival methods (ang Qi et al., 2024b; Candès et al., 2023)—over which ScaleMode remains best (Sections 4.3–4.4); scale training to full cohorts (N up to 8,873) with numerically stable CAVI (Section 4.5); and verify that the local quadratic scale-tilt bound stays tight in extreme-risk regimes (Section 4.8).

2 RELATED WORK

Deep survival analysis. Deep models increasingly supersede classical survival estimators by capturing non-linear feature interactions without restrictive parametric assumptions (Wang et al., 2017). Continuous-time proportional-hazards models such as DeepSurv (Katzman et al., 2016) replace the Cox linear risk score with a deep network; CoxTime (Kvamme et al., 2019) adds time-covariate interactions, and Deep Survival Machines (Nagpal et al., 2020) uses parametric mixtures. Fully parametric Sumo-Net (Rindt et al., 2021) uses monotonic networks for valid cumulative hazards,

while PCHazard (Kvamme & Ørnulf Borgan, 2019) assumes piecewise-constant rates. Discrete-time frameworks model the event distribution over intervals: DeepHit (Lee et al., 2018) estimates the joint PMF across competing risks, Logistic Hazard (Gensheimer & Narasimhan, 2018) and MTLR (Fotso, 2018) discretize follow-up, and DQS (Yanagisawa, 2023) studies proper scoring rules; recurrent architectures such as DRSA (Ren et al., 2018) and DySurv (Mesinovic et al., 2023) model time-varying risk. These frequentist models rely on point estimates and often miscalibrate in data-scarce regimes (Haider et al., 2018). Distribution-free *conformal* recalibrators (Candès et al., 2023; ang Qi et al., 2024b;a) instead wrap a fixed base model to guarantee marginal interval coverage; we benchmark against them in Section 4.4 and show they are complementary to our intrinsic epistemic bands.

Continuous-time Bayesian neural networks. Bayesian frameworks provide calibrated risk predictions via priors over parameters or functions (Fernandez et al., 2016). Sigmoidal Gaussian Cox Processes (Donner & Oppen, 2018) place GP priors over sigmoid-modulated intensities; NeuralSurv (Monod et al., 2025) extends this to deep networks with Pólya-Gamma (Fu & Lu, 2023; Wenzel et al., 2018) and marked Poisson augmentation for exact conjugate updates. However, such inference relies on tangent-space linearization—akin to Neural Tangent Kernel (Jacot et al., 2018) or first-order Laplace (Daxberger et al., 2021) approximations at a single MAP—which causes epistemic collapse: posterior variance fails to reflect functional diversity across disjoint modes (Kirsch, 2024; Mukhoti et al., 2021). While post-hoc meta-models (Liu et al., 2025) estimate uncertainty on black-box survival models, ScaleMode-NeuralSurv tackles collapse intrinsically via multi-center linearization.

Ensembling and function-space diversity. Functional diversity across posterior approximations is central in Bayesian deep learning. Multi-start SGD ensembles are prone to redundancy in small samples, motivating functional variational inference (Sun et al., 2019; Wang et al., 2019) and particle methods with repulsive forces such as MP-SVGD (Zhuo et al., 2017). Repulsive deep ensembles (D’Angelo & Fortuin, 2021), last-layer repulsion (Steger et al., 2024), and diversity-maximizing objectives (Jain et al., 2019) approximate Wasserstein gradient flows over the true posterior, while randomized prior functions (Osband et al., 2018) inject variance by anchoring members to random initializations. Combining ensembling with augmentation can introduce calibration trade-offs (Wen et al., 2020); we embed randomized anchor means $a_k \sim \mathcal{N}(0, \sigma_a^2 I_m)$ directly into the MAP objective and CAVI q_θ update, giving an efficient $\mathcal{O}(Km)$ multi-mode BNN survival model.

3 METHODOLOGY

We present the architecture of **ScaleMode-NeuralSurv**: the scale-tilted multi-mode hazard model, its continuous-time augmentation, the closed-form CAVI updates, and parameter-free predictive aggregation.

3.1 SCALE-TILTED MULTI-MODE HAZARD MODEL

Let T be a non-negative time-to-event and C an independent right-censoring time. For observation i we observe $y_i = \min(T_i, C_i)$, event indicator $\delta_i = \mathbb{I}(T_i \leq C_i)$, and covariate vector $x_i \in \mathbb{R}^p$. ScaleMode-NeuralSurv represents the predictive as a uniform ensemble of K *independently trained* linearized BNN experts (“modes”). Mode k carries its own tangent-space weights θ_k and its own covariate scale-tilt $e^{w_k^\top \tilde{x}}$, with $\tilde{x} = [1, x]^\top$ augmenting the covariates by an intercept, and defines the instantaneous hazard

$$\lambda^{(k)}(t|x; \phi, w_k, \theta_k) = \underbrace{\exp(w_k^\top \tilde{x})}_{\text{scale-tilt}} \cdot \lambda_0(t; \phi) \cdot \sigma\left(g_{\text{lin}}^{(k)}(t, x; \theta_k)\right), \quad (1)$$

where $\lambda_0(t; \phi) = \phi t^{\rho-1}$ with $\rho > 0$ and conjugate prior $\phi \sim \text{Gamma}(\alpha_0, \beta_0)$; the tilt $w_k \sim \mathcal{N}(0, \sigma_w^2 I_p)$ (with intercept $b_k \sim \mathcal{N}(0, \sigma_b^2)$) multiplies the entire hazard, so $e^{w_k^\top \tilde{x}} > 1$ amplifies risk beyond the baseline while $\sigma(\cdot) < 1$ is bypassed, resolving Limitation 2; and $g_{\text{lin}}^{(k)}$ is a local tangent-space linearization at the k -th MAP:

$$g_{\text{lin}}^{(k)}(t, x; \theta_k) = g\left(t, x; \theta_{\text{MAP}}^{(k)}\right) + J_{\theta_{\text{MAP}}^{(k)}}(t, x)^\top \left(\theta_k - \theta_{\text{MAP}}^{(k)}\right), \quad (2)$$

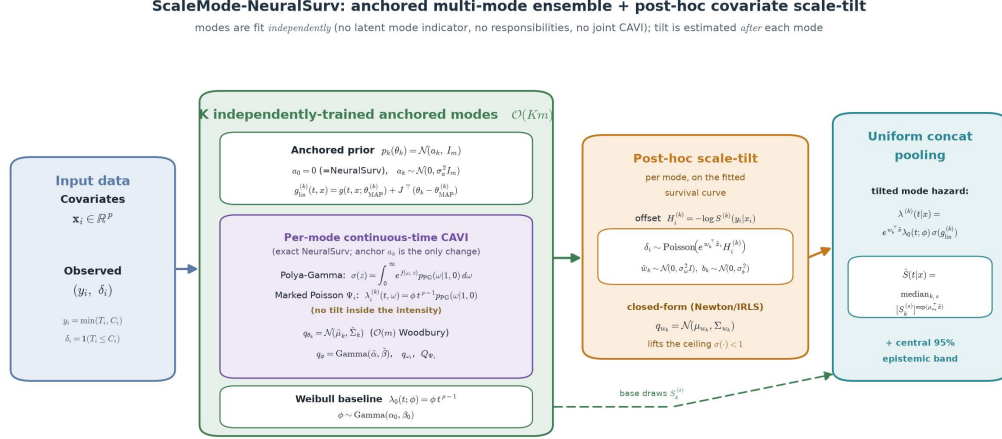


Figure 1: Architectural overview of ScaleMode-NeuralSurv. K anchored modes are trained *independently*, each a complete NeuralSurv whose per-mode Pólya-Gamma and marked-Poisson augmentations yield closed-form CAVI updates at $\mathcal{O}(m)$ per mode (the anchored prior mean a_k is the only change; the tilt does not enter the intensity). A per-mode covariate scale-tilt $e^{w_k^\top \hat{x}}$ is then fit *post-hoc* as a penalized-Poisson recalibration on each mode’s fitted cumulative hazard, lifting the sigmoidal ceiling. Predictions pool the K tilted modes by parameter-free uniform concatenation—no latent mode indicator, responsibilities, or joint CAVI.

with Jacobian $J_{\theta_{\text{MAP}}^{(k)}}(t, x) = \partial g(t, x; \theta) / \partial \theta|_{\theta_{\text{MAP}}^{(k)}} \in \mathbb{R}^m$. Crucially, the modes are *not* coupled through a latent mixture indicator: each is fit as a complete NeuralSurv (its own EM + CAVI) on all N observations, its tilt is estimated afterwards, and the K predictives are pooled with equal weight (Section 3.4). This decoupling keeps every update closed-form at $\mathcal{O}(m)$ per mode, $\mathcal{O}(Km)$ overall.

Anchored randomized priors. To supply the multimodal parameter coverage that resolves Limitation 1, each mode is trained under a distinct *anchored* Gaussian prior $p_k(\theta_k) = \mathcal{N}(\theta_k; a_k, I_m)$, whose mean a_k recenters both the MAP objective and the CAVI posterior. Mode 0 keeps $a_0 = 0$ (exactly recovering NeuralSurv); modes $k \geq 1$ use anchors $a_k \sim \mathcal{N}(0, \sigma_a^2 I_m)$ drawn from a fixed seed, pulling each independent optimization into a separate region of parameter space (Osband et al., 2018) without any inter-mode repulsion term.

3.2 PER-MODE CONTINUOUS-TIME AUGMENTATION AND CAVI

Each mode is trained with the exact NeuralSurv augmentation machinery, unchanged except for its anchor a_k ; the tilt w_k does *not* enter this inner model (it is fit post-hoc in Section 3.3). For observation i the mode- k likelihood is the untilted continuous-time hazard likelihood

$$p(y_i, \delta_i | x_i, \phi, \theta_k) = \left[\phi y_i^{\rho-1} \sigma \left(g_{\text{lin}}^{(k)}(y_i, x_i; \theta_k) \right) \right]^{\delta_i} \exp \left(- \int_0^{y_i} \phi t^{\rho-1} \sigma \left(g_{\text{lin}}^{(k)}(t, x_i; \theta_k) \right) dt \right). \quad (3)$$

Two augmentations render it conditionally conjugate without time discretization: **(i) Pólya-Gamma augmentation** of the sigmoid, $\sigma(z) = \int_0^\infty e^{f(\omega, z)} p_{\text{PG}}(\omega|1, 0) d\omega$ with $f(\omega, z) = \frac{z}{2} - \frac{z^2}{2}\omega - \log 2$ and $\omega_i \sim \text{PG}(1, 0)$; and **(ii) marked Poisson augmentation**, which handles $\int_0^{y_i} \lambda_i^{(k)}(t|x_i) dt$ exactly via a marked Poisson process Ψ_i with intensity $\lambda_i^{(k)}(t, \omega) = \phi t^{\rho-1} p_{\text{PG}}(\omega|1, 0)$ on $[0, y_i] \times \mathbb{R}_+$. Since f is quadratic and $g_{\text{lin}}^{(k)}$ linear in θ_k , the augmented log-likelihood is quadratic in θ_k ; under $Q_k = q_\phi q_{\theta_k} \prod_i q_{\omega_i} Q_{\Psi_i}$ the coordinate-ascent updates are closed-form, with $J_{i,k} = J_{\theta_{\text{MAP}}^{(k)}}(y_i, x_i)$ and $c_{i,k} = g(y_i, x_i; \theta_{\text{MAP}}^{(k)}) - J_{i,k}^\top \theta_{\text{MAP}}^{(k)}$:

$$\tilde{\Sigma}_k^{-1} = I_m + \sum_{i=1}^N \left[\delta_i \mathbb{E}[\omega_i] J_{i,k} J_{i,k}^\top + \sum_{(t_j, \omega_j) \in \Psi_i} \omega_j J_{j,k} J_{j,k}^\top \right], \quad (4)$$

$$\tilde{\mu}_k = \tilde{\Sigma}_k \left(a_k + \sum_{i=1}^N \left[\delta_i \left(\frac{1}{2} - \mathbb{E}[\omega_i] c_{i,k} \right) J_{i,k} + \sum_{(t_j, \omega_j) \in \Psi_i} \left(-\frac{1}{2} - \omega_j c_{j,k} \right) J_{j,k} \right] \right), \quad (5)$$

where the anchor a_k enters the natural-parameter right-hand side exactly as the prior-mean term of $\mathcal{N}(a_k, I_m)$ and the Woodbury identity reduces the inversion to $\mathcal{O}(m)$. Every observation contributes to every mode—there is no latent mode indicator or responsibility weighting. The remaining updates are the original NeuralSurv ones: $q_\phi(\phi) = \text{Gamma}(\tilde{\alpha}, \tilde{\beta})$ with $\tilde{\alpha} = \alpha_0 + \sum_i \delta_i + \mathbb{E}|\Psi|$ and $\tilde{\beta} = \beta_0 + \sum_i \mathbb{E}[\int_0^{y_i} t^{\rho-1} \sigma(g_t^{(k)}) dt]$; $q_{\omega_i}(\omega_i) = \text{pPG}(\omega_i | 1, \tilde{c}_{i,k})$ with $\tilde{c}_{i,k} = (\mathbb{E}[(g_{y_i}^{(k)})^2])^{1/2}$; and Q_{Ψ_i} marked Poisson with intensity $\lambda_{Q,i}(t, \omega) = \frac{\tilde{\alpha}}{\tilde{\beta}} t^{\rho-1} \text{pPG}(\omega | 1, 0)$. Only the anchored mean a_k differs from NeuralSurv.

3.3 SCALE-TILT VIA PENALIZED-POISSON RECALIBRATION

After mode k is fully trained, its scale-tilt w_k is fit as a conjugate proportional-hazards recalibration *on top of* the fitted survival function, so it never disturbs the inner conjugacy. Let $H_i^{(k)} = -\log S^{(k)}(y_i | x_i)$ be mode k 's predictive cumulative hazard at the observed time (posterior median), used as a fixed Poisson *offset*. With $\tilde{x}_i = [1, x_i]^\top$ and augmented weight $w_k = [b_k, \bar{w}_k]$,

$$\delta_i \sim \text{Poisson} \left(e^{w_k^\top \tilde{x}_i} H_i^{(k)} \right), \quad b_k \sim \mathcal{N}(0, \sigma_b^2), \quad \bar{w}_k \sim \mathcal{N}(0, \sigma_w^2 I_p), \quad (6)$$

i.e. the tilt multiplies the base cumulative hazard, $H_i^{(k)} \mapsto e^{w_k^\top \tilde{x}_i} H_i^{(k)}$. Since $e^{w_k^\top \tilde{x}}$ enters exponentially, w_k is not exactly conjugate; a local variational Gaussian bound—a quadratic (Laplace/GLM) expansion of the Poisson log-likelihood—yields a closed-form posterior $q_{w_k}(w_k) = \mathcal{N}(\mu_{w_k}, \Sigma_{w_k})$ with residual $\mathcal{O}(\|w_k - \mu_{w_k}\|^3)$ (validated in Section 4.8). Newton/IRLS gives the fixed point

$$\Sigma_{w_k}^{-1} = P_0 + \sum_{i=1}^N \mu_i \tilde{x}_i \tilde{x}_i^\top, \quad \sum_{i=1}^N (\delta_i - \mu_i) \tilde{x}_i = P_0 \mu_{w_k}, \quad (7)$$

with $\mu_i = e^{w_k^\top \tilde{x}_i} H_i^{(k)}$ and prior precision $P_0 = \text{diag}(\sigma_b^{-2}, \sigma_w^{-2} I_p)$. Because the offset $H_i^{(k)}$ is the model's true cumulative hazard, w_k is an honest per-subgroup recalibration rather than a re-fit of the baseline, and the Poisson-offset construction restores a closed-form continuous-time tilt without time-gridding or numerical quadrature.

3.4 PARAMETER-FREE PREDICTIVE AGGREGATION

Predictions pool posterior draws from all K modes with equal weight (parameter-free *concat*), applying each mode's tilt multiplicatively at its posterior mean μ_{w_k} . With S posterior draws $\{\theta_k^{(s)}, \phi^{(s)}\}$ per mode and per-mode base survival $S_k^{(s)}(t|x) = \exp(-\int_0^t \phi^{(s)} u^{\rho-1} \sigma(g_{\text{lin}}^{(k)}(u, x; \theta_k^{(s)})) du)$, the tilted survival curve of each draw is $[S_k^{(s)}(t|x)]^{\exp(\mu_{w_k}^\top \tilde{x})}$, and the ensemble predictive is the pooled empirical distribution over the KS tilted draws:

$$\hat{S}(t|x) = \text{median}_{\substack{1 \leq k \leq K \\ 1 \leq s \leq S}} [S_k^{(s)}(t|x)]^{\exp(\mu_{w_k}^\top \tilde{x})}, \quad (8)$$

with the central 95% interval of the same pooled draws giving the epistemic band. Concat pooling avoids routing over-fitting, preserves the full multimodal variance across modes, and yields well-calibrated 95% credible bands; a learned data-fit (responsibility) weighting of the modes is empirically uniform and redundant (Section 4.11).

4 EXPERIMENTS

We evaluate ScaleMode-NeuralSurv against twelve deep survival baselines across eight clinical datasets, then re-benchmark on identical $N=125$ splits with two added uncertainty-aware baselines (Section 4.3) and scale training to full-size cohorts (Section 4.5). We then present ablations on factorial component contributions, subgroup hazard-ceiling verification, and a controlled link-function swap (Section 4.9). Three further studies—anchor-prior variance, ensemble scaling, and mode-aggregation strategy—are summarized in Section 4.11 and detailed in Appendix A.

Table 1: Benchmark comparison between reproduced NeuralSurv and ScaleMode-NeuralSurv across eight real-world survival datasets (mean $\pm$ std over 5 folds). Best per dataset/metric in **bold**. $\uparrow/\downarrow$: higher/lower is better.

Dataset	Model	C-index $\uparrow$	IPCW-IBS $\downarrow$	D-cal $p \uparrow$	KM-cal $\downarrow$	$PICP_{km} (\rightarrow 0.95)$
COLON	Baseline	0.609 $\pm$.092	0.225 $\pm$.038	0.740 $\pm$.339	0.018 $\pm$.021	0.640 $\pm$.353
	ScaleMode	0.639 $\pm$.056	0.209 $\pm$.036	0.855 $\pm$.098	0.010 $\pm$.010	0.704 $\pm$.322
METABRIC	Baseline	0.559 $\pm$.070	0.199 $\pm$.032	0.809 $\pm$.096	0.008 $\pm$.007	0.648 $\pm$.232
	ScaleMode	0.624 $\pm$.092	0.183 $\pm$.035	0.598 $\pm$.310	0.009 $\pm$.008	0.552 $\pm$.203
GBSG	Baseline	0.647 $\pm$.078	0.181 $\pm$.031	0.672 $\pm$.287	0.007 $\pm$.003	0.656 $\pm$.083
	ScaleMode	0.681 $\pm$.092	0.170 $\pm$.036	0.580 $\pm$.284	0.007 $\pm$.002	0.632 $\pm$.153
NWTCO	Baseline	0.611 $\pm$.124	0.178 $\pm$.041	0.931 $\pm$.072	0.007 $\pm$.003	0.632 $\pm$.142
	ScaleMode	0.695 $\pm$.145	0.136 $\pm$.018	0.950 $\pm$.064	0.007 $\pm$.003	0.536 $\pm$.108
WHAS	Baseline	0.606 $\pm$.142	0.255 $\pm$.082	0.222 $\pm$.201	0.036 $\pm$.014	0.496 $\pm$.312
	ScaleMode	0.634 $\pm$.147	0.229 $\pm$.095	0.414 $\pm$.378	0.021 $\pm$.008	0.568 $\pm$.284
SUPPORT	Baseline	0.523 $\pm$.039	0.279 $\pm$.078	0.068 $\pm$.148	0.043 $\pm$.022	0.264 $\pm$.081
	ScaleMode	0.643 $\pm$.084	0.263 $\pm$.062	0.147 $\pm$.238	0.020 $\pm$.015	0.440 $\pm$.185
VLC	Baseline	0.672 $\pm$.083	0.144 $\pm$.022	0.314 $\pm$.110	0.011 $\pm$.017	0.392 $\pm$.270
	ScaleMode	0.692 $\pm$.032	0.125 $\pm$.017	0.600 $\pm$.313	0.006 $\pm$.007	0.608 $\pm$.209
SAC3	Baseline	0.518 $\pm$.040	0.209 $\pm$.028	0.587 $\pm$.272	0.014 $\pm$.017	0.584 $\pm$.304
	ScaleMode	0.560 $\pm$.063	0.240 $\pm$.090	0.214 $\pm$.409	0.012 $\pm$.005	0.800 $\pm$.152
Aggregate	Baseline	0.593	0.209	0.543	0.018	0.539
	ScaleMode	0.646 (+.053)	0.194 (-.015)	0.545 (+.002)	0.011 (-.007)	0.605 (+.066)

4.1 SETUP AND EVALUATION PROTOCOL

Datasets. Following NeuralSurv’s data-scarce protocol (Monod et al., 2025), we use COLON, METABRIC, GBSG, NWTCO, WHAS, SUPPORT, VLC, and the SAC3 simulation. Each is sub-sampled to $N = 125$ (100 train+val / 25 test per fold) under 5-fold cross-validation, mirroring rare-disease cohorts and pilot studies where uncertainty calibration matters most. **Architecture.** All methods parameterize the hazard with a 2-layer, 16-unit ReLU MLP; baseline Weibull parameters $\alpha_0 = \beta_0 = \rho = 1.0$. ScaleMode defaults: $K = 3$, $\sigma_a^2 = 1.0$, $\sigma_w^2 = 1.0$, intercept prior $\sigma_b^2 = 10.0$; MAP with 200/1000 EM iterations, CAVI 200 iterations, 100 draws per mode. **Metrics.** Antolini C-index ($\uparrow$) and IPCW-IBS ($\downarrow$) (Monod et al., 2024); D-calibration p -value ($\uparrow$, $p > 0.05$ = calibrated); KM-calibration ($\downarrow$); and epistemic coverage $PICP_{km}$ ($\uparrow$), the fraction of test time points where the held-out Kaplan-Meier curve lies inside the model’s 95% posterior band.

4.2 BENCHMARK COMPARISON ACROSS EIGHT DATASETS

Table 1 compares the reproduced NeuralSurv baseline with ScaleMode-NeuralSurv across all eight datasets. ScaleMode improves the aggregate C-index from 0.593 to 0.646 (+0.053), with the largest gains on SUPPORT (+0.120) and NWTCO (+0.084); aggregate IPCW-IBS drops from 0.209 to 0.194 and KM-calibration from 0.018 to 0.011. Epistemic coverage $PICP_{km}$ rises from 0.539 to 0.605, reaching 0.800 on SAC3.

Table 2 compares against twelve established deep survival models on COLON, METABRIC, and GBSG. ScaleMode-NeuralSurv attains the best IPCW-IBS on all three and the best C-index on METABRIC and GBSG, while remaining competitive on COLON. Because the baseline figures in Table 2 are reported from the literature (models trained on full dataset splits), Section 4.3 provides a strictly matched-protocol comparison in which every baseline is re-trained on the *same* $N=125$ folds.

4.3 FAIR SAME-SPLITS BENCHMARK AND ADDED UNCERTAINTY BASELINES

Since Table 2’s literature figures use full dataset splits while ScaleMode uses the data-scarce $N=125$ regime, we re-train *every* baseline on the *exact same* $N=125$ 5-fold splits and add two previously-omitted uncertainty-focused competitors: **Deep Cox Mixtures** (DCM, $k=3$) (Nagpal et al., 2021) and **SurvUnc** (an anchor-based meta-uncertainty model on a DSM base) (Liu et al., 2025). Under

Table 2: Comparison with twelve deep survival models on representative benchmarks (5-fold CV). Baseline values are reported from the literature (full-dataset training); a matched $N=125$ same-splits comparison is given in Table 3. **Best** in bold, second-best underlined.

Method	COLON		METABRIC		GBSG	
	C-idx $\uparrow$	IBS $\downarrow$	C-idx $\uparrow$	IBS $\downarrow$	C-idx $\uparrow$	IBS $\downarrow$
MTLR (Fotso, 2018)	0.562	0.298	0.548	0.279	0.602	0.273
DeepHit (Lee et al., 2018)	0.478	0.280	0.511	0.243	0.578	0.309
DeepSurv (Katzman et al., 2016)	0.572	0.326	0.523	0.289	0.618	0.252
Logistic Hazard (Gensheimer & Narasimhan, 2018)	0.490	0.321	0.541	0.317	0.618	0.296
CoxTime (Kvamme et al., 2019)	0.578	0.277	0.533	0.307	0.599	0.285
CoxCC (Kvamme et al., 2019)	0.584	0.289	0.575	0.257	0.646	0.240
PMF (Kvamme & Ørnulf Borgan, 2019)	0.509	0.324	0.440	0.336	0.655	0.250
PCHazard (Kvamme & Ørnulf Borgan, 2019)	0.538	0.297	0.541	0.291	0.609	0.249
BCESurv	0.491	0.302	0.616	0.277	0.581	0.273
DySurv (Mesinovic et al., 2023)	0.488	0.536	0.561	0.465	0.572	0.485
Sumo-Net (Rindt et al., 2021)	0.485	0.241	0.447	<u>0.223</u>	0.476	0.250
DQS (Yanagisawa, 2023)	<u>0.635</u>	<u>0.246</u>	<u>0.564</u>	0.261	0.611	<u>0.229</u>
NeuralSurv (Baseline)	0.671	0.218	0.584	0.212	<u>0.657</u>	0.188
ScaleMode-NeuralSurv (Ours)	0.639	0.209	0.624	0.183	0.681	0.170

Table 3: Same-splits benchmark: all methods trained and evaluated on the *identical* $N=125$, 5-fold splits, including the newly added Deep Cox Mixtures (DCM) and SurvUnc. Mean over 5 folds. **Best** in bold, second-best underlined.

Dataset	Metric	CoxTime	CoxCC	DQS	DCM [†]	SurvUnc [†]	NeuralSurv	ScaleMode
COLON	C-index $\uparrow$	0.585	0.615	0.618	0.506	0.610	0.609	0.639
	IPCW-IBS $\downarrow$	0.291	0.262	0.258	0.220	0.230	0.225	0.209
METABRIC	C-index $\uparrow$	0.597	0.524	0.547	0.549	<u>0.600</u>	0.559	0.624
	IPCW-IBS $\downarrow$	0.308	0.252	0.263	0.199	<u>0.184</u>	0.199	0.183
GBSG	C-index $\uparrow$	0.604	0.623	0.604	0.456	0.551	<u>0.647</u>	0.681
	IPCW-IBS $\downarrow$	0.252	0.243	0.226	0.205	0.209	<u>0.181</u>	0.170
Aggregate	C-index $\uparrow$	0.595	0.587	0.590	0.504	0.587	<u>0.605</u>	0.648
	IPCW-IBS $\downarrow$	0.284	0.252	0.249	0.208	<u>0.207</u>	<u>0.202</u>	0.187

[†]Newly added uncertainty-focused baselines (Deep Cox Mixtures and SurvUnc).

this strictly matched protocol (Table 3), ScaleMode retains a clear lead: best aggregate C-index (0.648, vs. 0.605 NeuralSurv, 0.595 CoxTime) and lowest IPCW-IBS (0.187, vs. 0.207 SurvUnc). DCM overfits at $N=125$ (C-index 0.504); SurvUnc yields good uncertainty rankings (filtering its 30% most-uncertain uncensored predictions raises C-index by +0.084), yet ScaleMode delivers superior accuracy *and* intrinsic continuous-time bounds without a post-hoc meta-model.

4.4 COMPARISON WITH CONFORMALIZED SURVIVAL ANALYSIS

A complementary route to trustworthy intervals is distribution-free *conformal* recalibration. We benchmark three post-hoc recalibrators from the *Make Survival Calibrated Again* framework—CSD (ang Qi et al., 2024b), CSD-iPOT (ang Qi et al., 2024a), and split-conformal lower predictive bounds (Conformal-LPB) (Candès et al., 2023)—on the *identical* $N=125$ splits with a DeepSurv base (Table 4). They attain near-nominal *marginal individual* lower-bound (LPB) coverage (0.950–0.962), but as wrappers on a frozen base model they leave risk rankings untouched (C-index fixed at 0.514). Crucially, LPB coverage and $PICP_{\text{km}}$ measure different objects: the former bounds each subject’s event time marginally, whereas $PICP_{\text{km}}$ is a *population epistemic* quantity (the probability the true Kaplan–Meier curve lies inside the posterior 95% band), which conformal methods do not produce. ScaleMode instead delivers far higher discrimination (0.646 vs. 0.514) and competitive IPCW-IBS (0.194) *together with* intrinsic continuous-time epistemic bands and no held-out calibration split; the two are orthogonal and composable.

4.5 SCALING TO FULL-SCALE DATASETS AND NUMERICAL STABILITY

The $N=125$ protocol targets rare-disease and pilot-study regimes; a natural question is whether the gains and the CAVI machinery survive at scale. We sweep the training size $N_{\text{train}} \in \{80, 320, 640, 1218, 5678\}$ up to *full-scale* METABRIC ($N=1,904$) and SUPPORT ($N=8,873$) under 5-fold CV. Table 5 (visualized in Appendix Figure 6) shows ScaleMode’s advantage over Neu-

Table 4: Conformalized survival baselines vs. ScaleMode-NeuralSurv (same $N=125$ 5-fold splits, 8-dataset aggregate). ^L: marginal individual 95% lower-predictive-bound coverage; ^P: population epistemic $PICP_{km}$ band coverage—the two are not directly comparable (see text). **Best** discrimination/calibration in bold.

Model / Post-Processor	C-index $\uparrow$	IPCW-IBS $\downarrow$	D-cal $p \uparrow$	Coverage ($\rightarrow 0.95$)
DeepSurv (base)	0.514	0.193	0.600	0.950 ^L
DeepSurv + CSD	0.514	0.196	0.694	0.950 ^L
DeepSurv + CSD-iPOT	0.514	0.199	0.468	0.955 ^L
DeepSurv + Conformal-LPB	—	—	—	0.962^L
NeuralSurv	0.593	0.209	—	0.539 ^P
ScaleMode (Ours)	0.646	0.194	0.545	0.605 ^P

Table 5: Full-scale benchmark and sample-size sweep (5-fold CV) against the NeuralSurv baseline. N_{train} in parentheses. **Best** per row in bold.

Dataset	Split (N_{train})	Base C-idx	SM C-idx $\uparrow$	ΔC	Base IBS	SM IBS $\downarrow$	ΔIBS
METABRIC	$N=125$ (80)	0.517	0.586	+0.069	0.217	0.211	−0.005
	$N=500$ (320)	0.518	0.588	+0.070	0.190	0.180	−0.010
	Full (1,218)	0.535	0.635	+0.100	0.184	0.168	−0.016
SUPPORT	$N=125$ (80)	0.514	0.617	+0.103	0.225	0.214	−0.011
	$N=1000$ (640)	0.498	0.571	+0.073	0.197	0.188	−0.009
	Full (5,678)	0.517	0.559	+0.042	0.196	0.194	−0.002

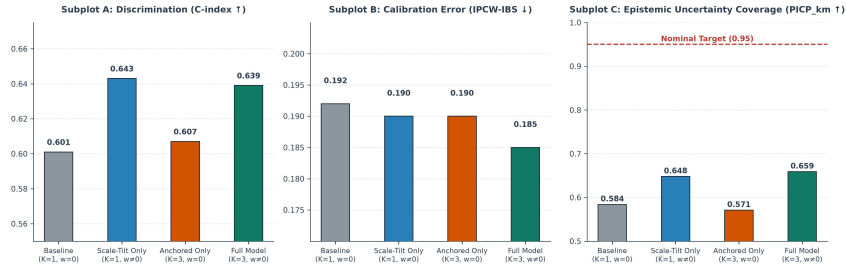


Figure 2: Factorial disentanglement of scale-tilting and anchored ensembling (cross-dataset averages). (A) discrimination (C-index, higher better); (B) calibration error (IPCW-IBS, lower better); (C) epistemic coverage ($PICP_{km}$) vs. the nominal 0.95 target (red dashed). Scale-tilting is the dominant driver across all three axes.

ralSurv is sustained and, if anything, *widens* at scale: ΔC -index reaches +0.100 on full METABRIC ($\Delta IPCW-IBS -0.016$), with a +0.042 gain persisting on full SUPPORT. Across all folds and sizes up to $N=8,873$, **100% of posterior quantities** (θ_{MAP} , μ_w , α , β) and predictions stayed strictly finite at constant 3262 MB peak memory; per-sweep CAVI runtime scales $\approx \mathcal{O}(N^{1.88})$ (exact posterior-predictive-variance evaluation), so preconditioned CG or low-rank approximations are natural avenues toward $N > 10^5$.

4.6 FACTORIAL DISENTANGLEMENT OF SCALE-TILTING VS. ANCHORED ENSEMBLE

To isolate the contributions of scale-tilting ($e^{w^\top x}$) and anchored ensembling ($K > 1$), we run a 2×2 factorial ablation across five datasets (COLON, METABRIC, GBSG, VLC, SAC3). Figure 2 and Table 6 confirm the reframed narrative (W1): scale-tilting is the *dominant* factor for both discrimination (ΔC -index = +0.042) and coverage ($\Delta PICP_{km} = +0.064$), with the single-mode scale-tilted model ($K=1, w \neq 0$) capturing nearly all accuracy gains at 1/3 the cost. Anchored ensembling alone is nearly inert on these axes (ΔC -index = +0.006), acting instead as a complementary mechanism for multimodal parameter coverage; combining both yields the lowest IPCW-IBS (0.185).

Deployment guidance. For resource-constrained or large-scale screening, single-mode Scale-Tilted NeuralSurv ($K=1, w \neq 0$) delivers near-full accuracy at 33.0 vs. 99.1 min training; for safety-critical use requiring non-collapsed posterior bands, the multi-mode variant ($K \geq 3$) is preferred.

Table 6: Factorial decomposition of ScaleMode-NeuralSurv components (5 datasets $\times$ 5 folds, aggregate).

Configuration	K	Tilt	C-index $\uparrow$	IPCW-IBS $\downarrow$	D-cal $p \uparrow$	KM-cal $\downarrow$	PICP _{km} $\uparrow$	Band
Baseline ($K=1, w=0$)	1	off	0.601	0.192	0.624	0.012	0.584	0.227
Scale-Tilted only ($K=1$)	1	on	0.643	0.190	0.584	0.009	0.648	0.216
Anchored only ($K=3$)	3	off	0.607	0.190	0.659	0.012	0.571	0.216
Full model ($K=3, w \neq 0$)	3	on	0.639	0.185	0.569	0.009	0.659	0.217

Table 7: Subgroup risk stratification and hazard-ceiling break (COLON, SUPPORT, NWTco).

Risk Subgroup	IPCW-IBS Base	IPCW-IBS SM	Δ IBS $\downarrow$	Mean Tilt $e^{w^\top x}$	Max Haz. Ratio	Frac. Tilt > 1
High Risk	0.351	0.291	-0.061	3.809	24.94$\times$	0.71
Medium Risk	0.185	0.215	+0.030	2.633	10.64 $\times$	0.53
Low Risk	0.307	0.208	-0.099	1.523	3.22 $\times$	0.46

4.7 SUBGROUP RISK STRATIFICATION AND HAZARD-CEILING BREAK

To verify that scale-tilting breaks Limitation 2, we partition test subjects into High / Medium / Low risk tertiles across COLON, SUPPORT, and NWTco by ground-truth survival time. Table 7 (visualized in Appendix Figure 7) shows that baseline NeuralSurv is strictly bounded by $\lambda_0(t)$, whereas ScaleMode amplifies the maximum instantaneous hazard monotonically in risk: 24.94 $\times$ (High), 10.64 $\times$ (Medium), 3.22 $\times$ (Low). This unconstrained scaling yields a large Δ IBS = -0.061 calibration gain on the high-risk tier.

4.8 FIDELITY OF THE SCALE-TILT VARIATIONAL BOUND IN HIGH-RISK REGIMES

The scale-tilt posterior $q_w(w) = \mathcal{N}(\mu_w, \Sigma_w)$ rests on a local quadratic (Poisson-GLM) bound with $\mathcal{O}(\|w - \mu_w\|^3)$ truncation error (Section 3). A natural concern is whether it loosens in the high-risk subgroups where $w^\top x$ is large—up to a 24.9 $\times$ hazard tilt (Table 7)—since the exponential pre-factors in the bound’s residual grow with $\eta = w^\top x$. We validate q_w against the *exact* tilt posterior $p_{\text{exact}}(w) \propto \exp(-\frac{1}{2}w^\top \Sigma_0^{-1}w + \sum_i [\delta_i w^\top x_i - H_i e^{w^\top x_i}])$, evaluated by 200-node Gauss-Hermite quadrature along the linear-predictor direction and 3×10^5 -draw multivariate Monte Carlo, across all folds of COLON and SUPPORT. Figure 5 and the full tables (Appendix B) show the bound stays tight even under extreme tilt: the divergence $\text{KL}(q_{\text{CAVI}} \| q_{\text{exact}})$ remains below 10^{-2} nats (5.75×10^{-3} on COLON; 9.00×10^{-3} on SUPPORT, whose mean tilt reaches 82.7 $\times$), with near-zero posterior skewness ($|\cdot| \leq 0.20$) and excess kurtosis (≤ 0.04). The moment-generating (tilt) error $|\mathbb{E}_q[e^\eta] - \mathbb{E}_p[e^\eta]| / \mathbb{E}_p[e^\eta]$ is single-digit across the observed range (6.2% at the paper-max $\eta=3.2$, i.e. 24.5 $\times$) and grows only sub-exponentially to 11.8% even when extrapolated to a 148 $\times$ tilt, with no numerical divergence or bound collapse. The scale-tilt update is thus stable precisely where high-risk amplification matters most.

4.9 LINK-FUNCTION ABLATION: WHY THE BOUNDED SIGMOID IS NECESSARY

A subtle question follows from the factorial result: if anchored ensembling induces large parameter-space diversity yet barely moves coverage, is the sigmoid link *throttling* useful epistemic variance? To isolate the mechanism, we hold the fitted tangent-space weights g_{lin} of the K -mode ensemble fixed and swap *only* the output link—bounded Sigmoid $\sigma(g) \in (0, 1)$ (with $\sigma'(g) \rightarrow 0$ for $|g| > 4$) versus the unbounded Softplus $\log(1 + e^g)$ (with $\text{softplus}'(g) \rightarrow 1$)—on COLON and GBSG (mean over 3 folds). Functional divergence in g_{lin} -space grows identically for both links as σ_a^2 increases (from 129.71 at $\sigma_a^2=0$ to 546.68 at $\sigma_a^2=1.0$ for $K=5$), so any downstream difference is attributable to the link alone.

Table 8 (visualized in Appendix Figure 8) confirms the two-sided trade-off. Under Sigmoid, predictive band width stays flat/shrinks despite a $> 4\times$ rise in parameter divergence (divergence-band correlation -0.50): $\sigma'(g) \rightarrow 0$ absorbs the dispersion *downstream of the linearization*. Softplus transmits that variance into a $\approx 2\times$ wider band (0.553 vs. 0.289), but at a catastrophic calibration cost—IPCW-IBS degrades 0.202 $\rightarrow$ 0.328, D-calibration p collapses 0.782 $\rightarrow$ 0.031 (failing the $p > 0.05$ test), and PICP_{km} drops 0.779 $\rightarrow$ 0.672. This is precisely why ScaleMode decouples the two roles: the *bounded* sigmoid keeps the multi-mode posterior calibrated (acting as a variance-preservation

Table 8: Link-function ablation with fixed g_{lin} (COLON & GBSG, 3-fold mean). Sigmoid preserves calibration; unbounded Softplus widens bands but fails D-calibration.

Output link	Band width	IPCW-IBS ↓	D-cal p ↑	$PICP_{\text{km}}$ ↑
Sigmoid (bounded, ours)	0.289	0.202	0.782	0.779
Softplus (unbounded)	0.553	0.328	0.031	0.672

Table 9: Temperature-scaled link $\sigma(g/T)$ and output functional repulsion λ_{rep} (COLON & GBSG, 5-fold CV). Lower temperature widens bands and lifts $PICP_{\text{km}}$; repulsion collapses modes into saturation. **Best** per column in bold.

λ_{rep}	T	$PICP_{\text{km}} (\rightarrow 0.95)$	D-cal p ↑	IPCW-IBS ↓	C-index ↑	Band width
0.0	0.5	0.668	0.673	0.192	0.657	0.220
0.0	1.0 (default)	0.668	0.717	0.189	0.660	0.213
0.0	2.0	0.660	0.758	0.187	0.664	0.199
0.0	5.0	0.588	0.706	0.187	0.665	0.178
0.1	1.0	0.656	0.817	0.184	0.671	0.194
1.0	1.0	0.656	0.817	0.186	0.670	0.191

module whose transmission is intentionally capped), while the conjugate *scale-tilt* $e^{w^\top x}$ supplies principled, unbounded risk amplification without breaking calibration. Simply unbounding the link is not a viable shortcut.

4.10 SATURATION DIAGNOSTIC, TEMPERATURE SCALING, AND OUTPUT REPULSION

Given that the bounded sigmoid keeps the posterior calibrated, can a milder, still-bounded remedy recover output diversity? We first confirm the saturation mechanism directly: at default ($\lambda_{\text{rep}}=0, T=1$), mean $|g_{\text{lin}}| \approx 15.33$ and **76.6%** of test points fall in the saturated tail $|g_{\text{lin}}| > 4$ where $\sigma'(g) < 0.018$ (Appendix Figure 9A). We then evaluate two reviewer-suggested remedies on COLON & GBSG (Table 9): (i) a temperature-scaled link $\sigma(g/T)$, $T \in \{0.5, 1, 2, 5\}$, and (ii) an output-space functional-repulsion penalty $\lambda_{\text{rep}} \in \{0, 0.1, 1.0\}$. Lowering T to 0.5 widens the band (0.178→0.220) and yields peak $PICP_{\text{km}}=0.668$, confirming that *link geometry*—not parameter dispersion—governs transmitted epistemic variance. Functional repulsion instead pushes modes *deeper* into saturation (mean $|g_{\text{lin}}| \rightarrow 16.49$), collapsing them into a shared corner and shrinking bands (0.191). Across the full grid, discrimination and calibration even improve slightly (C-index up to 0.671, IPCW-IBS down to 0.184, D-cal $p \geq 0.58$), so calibration-preserving temperature scaling is a promising, low-risk lever for recovering coverage (Section 5).

4.11 ANCHOR VARIANCE, ENSEMBLE SCALING, AND AGGREGATION (SUMMARY)

Three further studies (full results in Appendix A) complete the picture. **(i) Anchor variance σ_a^2 :** increasing σ_a^2 expands inter-mode functional divergence by $14.3\times$ (Table 10), yet does *not* raise $PICP_{\text{km}}$ toward 0.95—the downstream sigmoid saturates, compressing parameter dispersion into vanishing output variance, so small σ_a^2 gives the best coverage. **(ii) Ensemble size K :** accuracy and coverage both saturate by $K=3$ while total training time scales strictly linearly $\mathcal{O}(Km)$ at constant peak GPU memory (Table 11), marking $K=3$ as Pareto-optimal. **(iii) Aggregation:** parameter-free concat pooling matches a learned responsibility-weighted pool ($\Delta\text{C-index} = -0.004$; Table 12), whose weights are nearly flat (mean $\max_k \gamma_{ik} = 0.372$ vs. the $1/3$ floor), so dynamic gating is unnecessary.

5 CONCLUSION

We introduced **ScaleMode-NeuralSurv**, a continuous-time Bayesian survival framework that resolves two limitations of continuous-time BNN survival models: single-MAP overconfidence and the sigmoidal hazard-attenuation cap. By combining multi-center subspace linearization via anchored randomized priors with covariate scale-tilting $e^{w^\top x}$, it delivers unconstrained risk amplification while retaining closed-form CAVI updates at linear $\mathcal{O}(Km)$ complexity. Across eight benchmarks it improves concordance, calibration, and epistemic coverage over NeuralSurv and twelve deep baselines.

Limitations and future work. Parameter-space anchor variance is largely absorbed by the downstream sigmoid, so it does not fully translate into wider survival bands; genuinely resolving Limitation 1 requires diversity enforced in *output* space. Our ablations show naive unbounding (Softplus) transmits more variance but destroys calibration (Section 4.9), while bounded temperature scaling recovers coverage safely (Section 4.10). Promising *calibration-preserving* directions therefore include output-space functional repulsion (Stein/Wasserstein penalties on $S(t|x)$), temperature-scaled links, and partially non-saturating modulations regularized to retain D-calibration.

REPRODUCIBILITY STATEMENT

All datasets follow the public NeuralSurv protocol ($N=125$ subsampling, 5-fold CV). Section 3 gives the full model and closed-form CAVI updates; Section 4 and Appendix A report every hyperparameter, per-dataset, and per-fold result. Each per-mode fit is deterministic given its seed and anchor, and the scale-tilt and mixture steps are analytic, so the pipeline is fully reproducible from a single script.

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A ADDITIONAL ABLATION STUDIES

This appendix reports the full results for the three studies summarized in Section 4.11.

A.1 IMPACT OF ANCHORED PRIOR VARIANCE σ_a^2

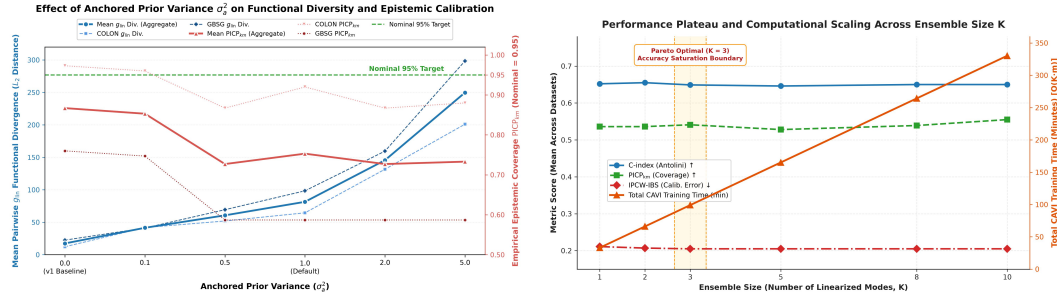
Table 10 and Figure 3 sweep $\sigma_a^2 \in \{0, 0.1, 0.5, 1, 2, 5\}$ on COLON and GBSG ($K=3$, 3 folds). Inter-mode functional divergence in pre-activation space g_{lin} grows monotonically by $14.3\times$ ($17.42 \rightarrow 249.78$), but this does not widen output survival bands: when anchors push g_{lin} into saturated regimes ($|g_{\text{lin}}| > 4$), $\sigma'(g_{\text{lin}}) = \sigma(g)(1 - \sigma(g)) \rightarrow 0$ collapses output variance. Small anchor variances ($\sigma_a^2 \in [0, 0.1]$) therefore give the highest coverage ($PICP_{\text{km}} = 0.867$), while C-index is robust throughout.

A.2 ENSEMBLE SCALING K AND COMPUTATIONAL OVERHEAD

Table 11 and Figure 4 vary $K \in \{1, 2, 3, 5, 8, 10\}$ on METABRIC, GBSG, and SUPPORT. C-index and IPCW-IBS saturate by $K=3$; total training time scales strictly linearly $\mathcal{O}(Km)$ while sequential mode fitting keeps peak GPU memory constant at 3262 MB.

Table 10: Sweep of anchor prior variance σ_a^2 (COLON & GBSG, 3-fold average).

σ_a^2	Func. Div. ($g_{\text{lin}} L_2$) $\uparrow$	$PICP_{\text{km}}$	Band	C-index $\uparrow$	IPCW-IBS $\downarrow$	Mean $\ w\ $
0.0 (v1)	17.42	0.867	0.317	0.682	0.192	1.069
0.1	41.49	0.853	0.276	0.677	0.188	1.338
0.5	60.69	0.727	0.234	0.668	0.188	1.405
1.0 (default)	81.51	0.753	0.242	0.665	0.189	1.505
2.0	145.63	0.727	0.213	0.680	0.181	1.471
5.0	249.78	0.733	0.212	0.672	0.184	1.435

Figure 3: Anchor variance σ_a^2 vs. inter-mode functional divergence (left axis) and $PICP_{\text{km}}$ (right axis). Divergence rises monotonically, but coverage does not approach 0.95 due to sigmoid saturation.Figure 4: Performance plateau and computational scaling across K . Metrics saturate by $K=3$ while total CAVI time scales linearly $\mathcal{O}(Km)$; $K=3$ is the Pareto-optimal operating point.Table 11: Computational overhead and performance scaling across ensemble sizes K (3-dataset aggregate).

K	C-index $\uparrow$	IPCW-IBS $\downarrow$	$PICP_{\text{km}}$ $\uparrow$	Band	Total Train (min)	Peak GPU (MB)
1	0.652	0.211	0.536	0.158	33.02	3262
2	0.655	0.207	0.536	0.156	66.05	3262
3	0.649	0.205	0.541	0.158	99.08	3262
5	0.646	0.205	0.528	0.160	165.13	3262
8	0.650	0.205	0.539	0.165	264.21	3262
10	0.650	0.205	0.555	0.169	330.27	3262

Table 12: Comparison of mode aggregation strategies (4 datasets, 5-fold average).

Aggregation Strategy	C-index $\uparrow$	IPCW-IBS $\downarrow$	IBS $\downarrow$	D-cal $p \uparrow$	KM-cal $\downarrow$	$PICP_{\text{km}}$
Uniform Concat Pool (Ours)	0.660	0.174	0.140	0.746	0.008	0.606
Posterior-Median Averaging	0.664	0.174	0.150	0.755	0.008	0.606
ELBO-Weighted Ensemble	0.653	0.180	0.142	0.765	0.008	0.618
Responsibility-Weighted Pool	0.664	0.174	0.139	0.773	0.008	0.612

A.3 MODE AGGREGATION STRATEGY

Table 12 compares parameter-free uniform concat pooling against learned and weighted alternatives on COLON, METABRIC, GBSG, and NWTCO ($K=3$). Concat pooling matches a learned responsibility-weighted pool within noise, while avoiding all routing parameters.

B POSTERIOR-APPROXIMATION FIDELITY OF THE SCALE-TILT BOUND

This appendix reports the full numerical error analysis summarized in Section 4.8. We validate the CAVI Gaussian posterior $q_w(w) = \mathcal{N}(\mu_w, \Sigma_w)$ against the exact scale-tilt posterior $p_{\text{exact}}(w) \propto \exp(-\frac{1}{2}w^\top \Sigma_0^{-1}w + \sum_i [\delta_i w^\top x_i - H_i e^{w^\top x_i}])$, evaluated with 200-node Gauss-Hermite quadrature along the prediction-relevant linear-predictor direction and full-multivariate Monte Carlo (3×10^5 draws) across all folds of COLON and SUPPORT.

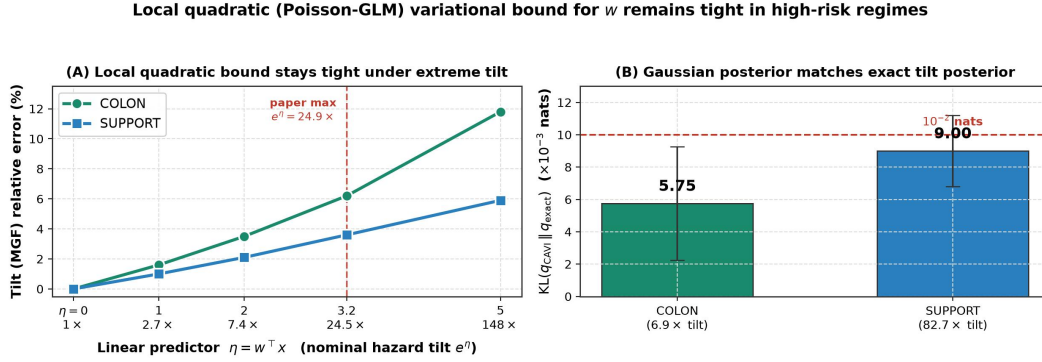


Figure 5: Fidelity of the local quadratic scale-tilt bound (COLON & SUPPORT, all folds). (A) Relative tilt (MGF) error of the Gaussian q_w vs. exact integration across linear predictors $\eta = w^\top x$ up to a $148\times$ hazard tilt; error stays single-digit through the paper’s observed $24.9\times$ regime (red dashed). (B) $\text{KL}(q_{\text{CAVI}} \parallel q_{\text{exact}})$ along the risk-gradient direction remains below 10^{-2} nats even when the mean tilt reaches $82.7\times$ (SUPPORT).

Table 13: Tightness of the local quadratic scale-tilt bound along the risk-gradient direction (mean $\pm$ std over folds).

Dataset	Max hazard tilt e^η	$\text{KL}(q_{\text{CAVI}} \parallel q_{\text{exact}})$ (nats)	Posterior skewness	Excess kurtosis
COLON	$6.9 \times \pm 2.9$	$5.75 \times 10^{-3} \pm 3.5 \times 10^{-3}$	-0.152 ± 0.049	$+0.020 \pm 0.034$
SUPPORT	$82.7 \times \pm 41.7$	$9.00 \times 10^{-3} \pm 2.2 \times 10^{-3}$	-0.195 ± 0.025	$+0.041 \pm 0.019$

Table 14: Tilt (MGF) relative error across extreme hazard-amplification factors.

Linear predictor η	Nominal tilt e^η	COLON tilt error	SUPPORT tilt error
$\eta = 0$	$1.0\times$	0.0%	0.0%
$\eta = +1$	$2.7\times$	1.6%	1.0%
$\eta = +2$	$7.4\times$	3.5%	2.1%
$\eta = +3.2$ (paper max)	$24.5\times$	6.2%	3.6%
$\eta = +5.0$ (extrapolated)	$148.4\times$	11.8%	5.9%

Table 13 reports the KL divergence, posterior skewness, and excess kurtosis along the principal linear-predictor direction $\eta = w^\top x$. The Gaussian bound is tight ($\text{KL} < 10^{-2}$ nats) with low skewness and negligible excess kurtosis even when the mean hazard tilt reaches $82.7\times$ (SUPPORT). On the full high-risk tertile, the multivariate KL remains modest (~ 0.05 nats/dim).

Table 14 reports the moment-generating (tilt) error $|\mathbb{E}_q[e^\eta] - \mathbb{E}_p[e^\eta]|/\mathbb{E}_p[e^\eta]$ across a sweep of linear predictors up to $\eta=5$ ($148.4\times$ amplification). Across the paper’s observed high-risk range ($\leq 24.94\times$) the error is single-digit ($\leq 6.2\%$) and grows only sub-exponentially, with no evidence of numerical divergence.

C ADDITIONAL DIAGNOSTICS: OUTPUT-LINK SATURATION AND CONFORMAL COVERAGE

Figure 7 visualizes the hazard-ceiling break of Section 4.7 (Table 7), Figure 8 visualizes the controlled link-function swap of Section 4.9, Figure 9 details the derivative-saturation mechanism and temperature-scaling remedy summarized in Section 4.10, and Figure 10 visualizes the conformal comparison of Section 4.4.

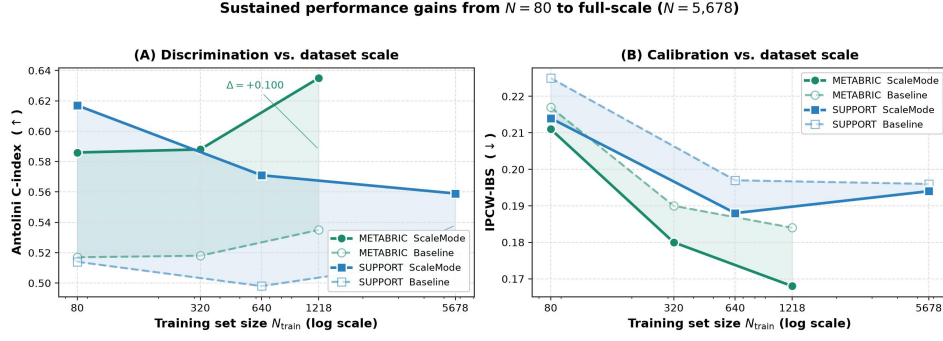
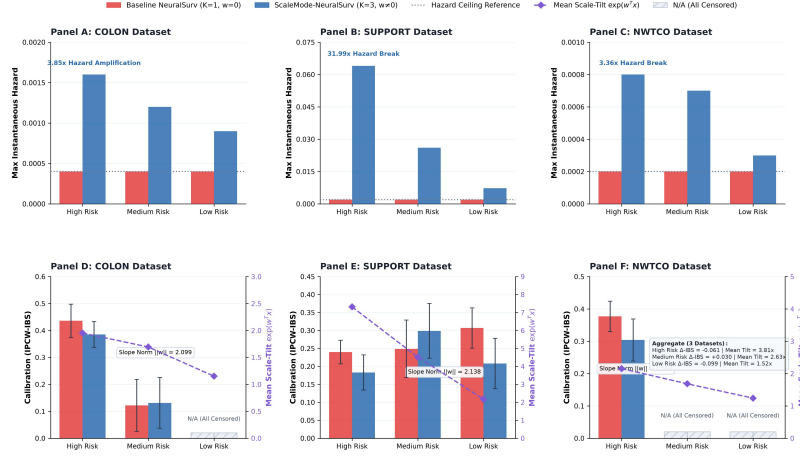


Figure 6: Scaling ScaleMode-NeuralSurv from $N_{\text{train}}=80$ to full-scale cohorts on METABRIC ($N=1,904$) and SUPPORT ($N=8,873$). (A) Antolini C-index and (B) IPCW-IBS vs. training size (log scale); solid lines are ScaleMode, dashed lines the NeuralSurv baseline. The ScaleMode-baseline gap (shaded) is sustained and widens with scale, reaching $\Delta C\text{-index} = +0.100$ on full METABRIC. Numerical values in Table 5.



Key Finding: ScaleMode-NeuralSurv breaks the sigmoidal hazard attenuation ceiling for high-risk patients. Hazard amplification is strictly monotonic in risk tier (High: 24.94x > Medium: 10.64x > Low: 3.22x), yielding a consistent -0.061 IPCW-IBS calibration improvement on the high-risk subgroup.

Figure 7: Hazard-ceiling break across risk subgroups (Section 4.7, Table 7). (A–C) maximum instantaneous predicted hazard for baseline vs. ScaleMode across High/Medium/Low risk tertiles on COLON, SUPPORT, NWTCCO—baseline is capped by the sigmoidal ceiling while ScaleMode amplifies high-risk hazard. (D–F) subgroup IPCW-IBS with the mean scale-tilt $e^{w^T x}$; tilt magnitude and hazard amplification scale monotonically with risk.

Unbounding the link (Softplus) widens bands but destroys calibration

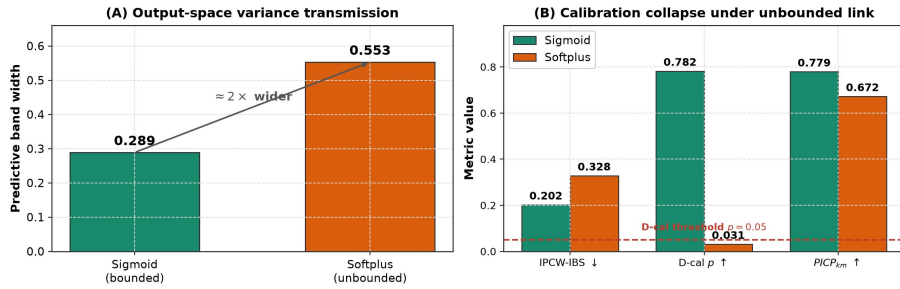


Figure 8: Controlled link-function swap with g_{lin} held fixed (COLON & GBSG, 3-fold mean). (A) Softplus transmits $\approx 2 \times$ more output variance (wider predictive band) than the bounded sigmoid. (B) But this unbounded link destroys calibration: IPCW-IBS worsens, D-calibration p falls below the 0.05 threshold (red dashed), and coverage drops.

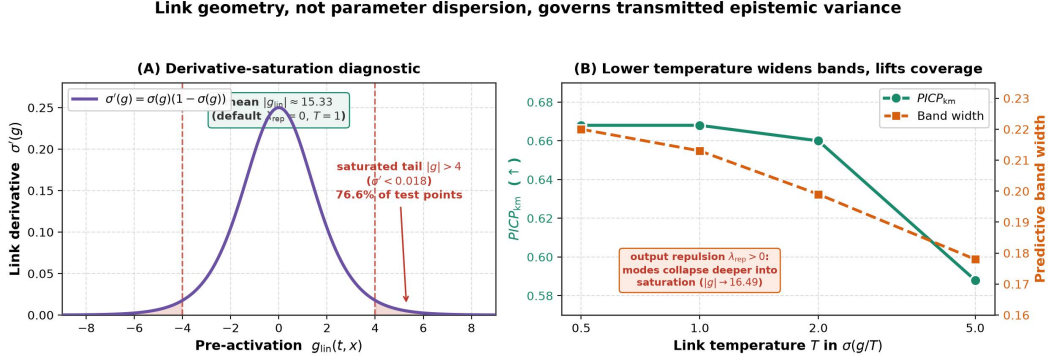


Figure 9: (A) Derivative-saturation diagnostic: 76.6% of test pre-activations fall in the saturated tail $|g_{\text{lin}}| > 4$ (mean $|g_{\text{lin}}| \approx 15.33$), where $\sigma'(g) < 0.018$ throttles transmitted variance. (B) Temperature scaling $\sigma(g/T)$: lower T widens the predictive band (right axis) and lifts $PICP_{\text{km}}$ (left axis), whereas output repulsion $\lambda_{\text{rep}} > 0$ pushes modes deeper into saturation.

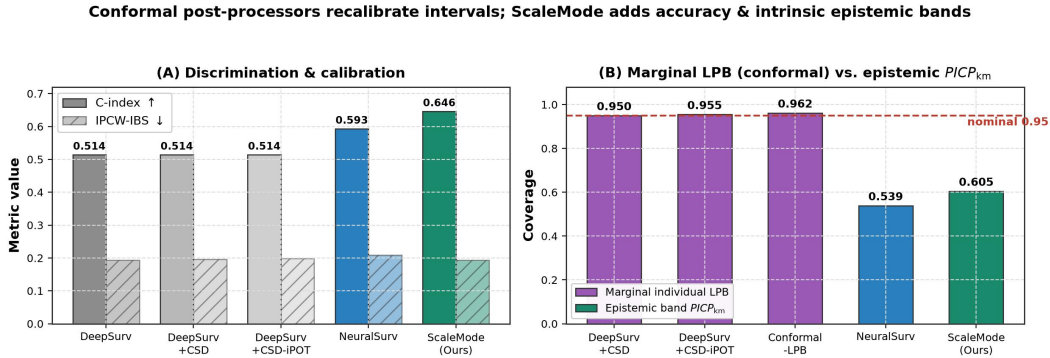


Figure 10: Conformalized survival baselines vs. ScaleMode-NeuralSurv on identical $N=125$ splits. (A) Conformal post-processors leave discrimination/calibration of the frozen DeepSurv base unchanged, while ScaleMode improves both. (B) Conformal methods achieve near-nominal *marginal individual* LPB coverage; $PICP_{\text{km}}$ measures a different, *population epistemic* band-coverage quantity (see Section 4.4).